Tong Zhao

Curriculum Vitae

+086 19975519969
zhaotong@ict.ac.cn

Position

2021-Current **Research Associate**, *Institute of Computing Technology Chinese Academy of Science*, Beijing, China.

Education

2016–2021 **PhD Student of Stochastic Partial Differential Equation**, *Department of Stochastic Control and Operation Research of Fudan University*, Shanghai, China.

Supervisor Shanjian Tang

Knowledge Control Theory & Optimal Control Theory, Stochastic Control Theory, PDE, SDE, SPDE, BSDE, Dynamic System, Neural Network, Deep Learning, Mathematical Statistics, Finite Elements Method, Sobolev Space, Functional Analysis, Stochastic Analysis, etc.

2012–2016 **Bachelor of Mathematics**, *School of Mathematical Science of Fudan University*, Shanghai, China.

Numerical Analysis, Numercial Solution of Differential Equations, Variation Method, Group Representation, Set Theory, Topology, Probability Theory, Algebric Topology, Reimann Geometry, Abstract Algebra, Physics&Mathematical Equations, Generalized Functions, Fourier Analysis, Real Analysis, Complex Analysis, Mathematical Analysis, Linear Algebra, Analytic Geometry, Mechanics, Measure Theory.

Article

Title Limits of one-dimensional interacting particle systems with two-scale interaction

Abstract This research, based on interacting partical system, characterizes the limits of a large system of interacting particles distributed on the real line. The interaction occurring among neighbors involves two kinds of independent actions with different rates. Besides this, the rates of these interactions can be state dependent, which means these rates depend on the distribution of the two kinds particals. This system is a generalization of the voter process and contact process, of which each particle is of type **A** or a. Under suitable scaling, the local proportion functions of **A** particles converge to continuous functions which solve a class of stochastic partial differential equations driven by Fisher--Wright white noise or branching white noise. To obtain the convergence, the tightness of these functions are derived from the moment estimate method.

Programming

Title Solving fully-coupled FBSDEs and corresponding quasi-linear PDEs with deep-learning

Abstract I developed 4 deep-learning based algorithms to solve fully-coupled FBSDEs and corresponding quasi-linear PDEs, which performed well to tackel with curse of dimension. I implement it with python and tensorflow, and then choose a FBSDE which has a explicit solution to check the relative error of my algorithm. In the worse case, the relative error is 0.5%.

Academic Experience

August 2018 **International Congress of Mathematicians**, *Institute of Pure and Applied Mathematics*, Rio de Janeiro, Brazil.
Attending as a school-awarded participant.

July 2018 **Riemann International School of Mathematics**, *Riemann International School*, Varese, Italy.
Attending as an awarded PhD student. Lecturer: Martin Hairer

July 2018 **Conference on Regularity Structure and Stochastic System**, *Chinese Academy of Science*, Beijing, China.
Attending as an awarded PhD student. Lecturer: Martin Hairer

June 2018 **Sino-U.S. Economists Forum**, *Fudan University*, Shanghai, China.
Attending as an PhD student of my own school. Lecturer: Darrell Duffie, Shige Peng.

June 2018 **The joint International Meeting of Chinese Mathematical Society & American Mathematical Society**, *Fudan University*, Shanghai, China.
Attending as an PhD student of my own school.

April 2018 **Workshop on Markov Process**, *Southern University of Science and Technology*, Shenzhen, China.
Attending as a Shanghai-awarded participant. Lecturer: Jie Xiong.

January 2018 **Workshop on Stochastic Partial Differential Equation and Related Field**, *Beijing Jiaotong University*, Beijing, China.
Attending as a school-awarded participant. Lecturer: Jiming Ma.

July 2017 **Summer School of Financial Engineering**, *Shangdong University*, Weihai, China.
Attending as a school-awarded participant. Lecturer: Gechun Liang, Shanjian Tang.

Computer Skills

OS	Mac OS, Windows	Image	PS, AI, Asymptote
Coding	Python(Tensorflow, PyTorch), LaTeX, Matlab	Website	HTML, CSS